

AI-Powered Learning & Development at Tata Steel

Transforming workforce learning through AI-driven Intelligence

Problem Statement & Business Case

"When an expert retires, sometimes an entire library leaves with them."

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THE L&D LANDSCAPE AT TATA STEEL

 **80,000+**

Employees Worldwide

~65,000 in India alone

 **3 Types**

Workforce Categories

Shopfloor · Engineers · Managers

 **100s**

Training Programs/Year

Safety, Skills, Leadership & more

  **High Risk**

Knowledge Loss

Senior workers retiring every year

Understanding the Context

- Tata Steel manages a large and diverse workforce across plants, mines, and offices.
- Training needs differ significantly between shopfloor workers, engineers, and managers.
- However, most current training programs are still standardized and classroom-driven, making personalization and skill tracking difficult at scale.

KEY CHALLENGES IN L&D TODAY

1. Lack of Personalization 🎯

- Same training for all employees.
- No role-based learning.
- Individual skill gaps remain hidden.



2. Low Training Effectiveness 📉

- Scheduled classroom sessions.
- Static PDFs lead to poor retention.
- Completion rates don't reflect actual learning or skill development.



3. Invisible Skill Gaps 📊

- No real-time skill tracking.
- Difficult to identify workforce readiness.
- Problems detected only after operational impact.



4. Knowledge Loss from Retirements 🤔

- Senior employees with 20-30 years of tacit knowledge.
- Limited documentation of practical experience.
- Years of shopfloor experience leave with retiring workers.



PRIMARY PROBLEM STATEMENT

Shopfloor workers and new joiners at Tata Steel lack personalized, on-demand training aligned to their individual skill gaps. Current training is scheduled, generic, and one-size-fits-all — leaving critical learning needs unmet at scale.



WHO IS AFFECTED?

- Shopfloor operator.
- Technicians.
- New joiners.
- Operational workforce.



WHAT IS THE PROBLEM?

- No personalized learning experience.
- Limited real-time access to knowledge.
- Same training for employees with different skill levels.



CURRENT GAP

- Classroom-heavy learning model.
- Static training content.
- No adaptive learning system



BUSINESS IMPACT

- Slower onboarding.
- Reduced productivity.
- Higher operational & safety risk.
- Low training ROI.

SECONDARY OPPORTUNITY — KNOWLEDGE PRESERVATION

"When an expert retires, sometimes an entire library leaves with them."

The Hidden Risk ⚠️



Tata Steel employs thousands of senior workers with decades of hands-on experience.

- Much of this knowledge is undocumented.
- Retirement leads to permanent knowledge loss .
- New employees cannot easily access real-world experience.

When they retire or leave, this operational intelligence is permanently lost.



The AI Opportunity ?



AI can systematically capture expert knowledge through:

- Capture expert insights through interviews & Q&A.
- Digitize SOPs, manuals, and troubleshooting methods.
- Build searchable AI knowledge systems.
- Enable 24/7 access to operational knowledge.

This turns individual expertise into organizational intelligence.

"AI enables Tata Steel to transform individual experience into organizational intelligence."

BUSINESS IMPACT OF INEFFECTIVE L&D

 **Safety**

“Poor training increases operational errors and workplace safety risks”

CRITICAL

 **Productivity**

“Slower onboarding reduces workforce efficiency and operational productivity”

HIGH

 **Training ROI**

“Generic training programs have poor retention rates and limited measurable skill improvement”

HIGH

 **Knowledge Capital**

“Critical operational expertise is lost when experienced employees retire”

LONG-TERM

“Ineffective learning systems directly impact safety, productivity, and operational continuity.”

WHY AI IS THE RIGHT SOLUTION

Scale

AI can train 65,000 employees simultaneously — impossible for any human trainer team.

Personalization

AI adapts to each individual's skill level, role, and learning pace in real time.

24/7 Availability

Workers can access knowledge and training support anytime, including during shifts.

Knowledge Capture

AI can systematically collect and organize expert knowledge before it is lost.

Industry Adoption

Siemens

AI chatbot for shopfloor training queries

IBM

AI-powered personalized learning paths

Unilever

AI skill gap assessment at scale

Infosys

GenAI for employee upskilling

Industry Avg.

40% improvement in training efficiency with AI L&D tools

“AI transforms learning from reactive training into continuous intelligent development”

SOLUTION DESIGN

TataBot: AI Knowledge Assistant for Learning & Development

From Problem Identification → AI Solution Blueprint

Designing TataBot to address workforce learning, knowledge access and skill development challenges

"Delivering the right knowledge to the right employee at the right time."

USER PERSONAS — WHO WILL USE TATABOT?



Ramesh, 22

New Shopfloor Operator

Joined 2 weeks ago

Challenge:

- New to operations
- Large training manuals
- Limited supervisor availability

Goal:

- Learn safety procedures
- Perform daily tasks confidently

"I don't know who to ask when my supervisor is busy."



Suresh, 31

Maintenance Technician

5 years experience

Challenge :

- Finds SOPs slowly
- Downtime during troubleshooting

Goal:

- Resolve issues faster
- Reduce dependency on supervisors

"By the time I find the right SOP page, the machine has been down 20 minutes."



Priya Sharma, 40

L&D Manager

Corporate Training Dept.

Challenge :

- Limited visibility into learning effectiveness
- Skill gaps difficult to track

Goal:

- Improve training outcomes
- Monitor workforce capability

"I have no idea if last month's training actually changed anything on the ground."

USER JOURNEY — RAMESH (NEW JOINER)

✗ TODAY — Without AI

? Ramesh has a safety-related question



👤 Supervisor is busy on shopfloor



📖 Ramesh searches large training manual



🔍 Cannot identify correct procedure quickly



😞 Feels confused and hesitant



⚠️ Risk of incorrect operation or safety issue

✓ WITH TATABOT

📱 Ramesh opens TataBot on mobile/tablet



💬 Asks: "What PPE is required in furnace area?"



📄 AI gives instant simplified guidance



📋 AI also explains safety precautions step by step



🎯 AI asks: "New to this area? Want a safety checklist?"



✓ Ramesh follows correct process confidently

TataBot reduces learning friction by providing real-time, role-specific guidance

USER JOURNEY — SURESH (EXPERIENCED TECHNICIAN)

✗ TODAY — Without AI

🔍 Machine issue occurs during operation



📖 Suresh searches multiple SOP documents



🔍 Updated info is difficult to locate quickly



👤🏭 Contacts supervisor or senior engineer



🔍 Production delay increases



⚠️ Machine downtime impacts operations

✓ WITH TATABOT

📱 Suresh opens TataBot on device



💬 Types: "Error code E-47 on rolling mill"



🔍 AI searches SOPs and troubleshooting database



📋 AI provides step-by-step resolution instantly



🔧 AI asks: "Log this issue for maintenance records?"



✓ Reduced downtime, faster decision-making

TataBot accelerates troubleshooting by providing instant access to relevant operational knowledge

USER JOURNEY — PRIYA (L&D MANAGER)

✗ TODAY — Without AI

 Training sessions are conducted manually



 Employee engagement is difficult to measure



 Skill gaps are not clearly identified



 Feedback collection takes time



 No real-time visibility into effectiveness



 Training impact remains uncertain

✓ WITH TATABOT

 Priya accesses AI-powered training dashboard



 AI tracks all employee learning interactions



 Skill gaps and trends analyzed automatically



 AI recommends personalized learning paths



 Training effectiveness becomes measurable



✓ Priya presents data-backed report to leadership

TataBot enables proactive learning decisions through real-time analytics and workforce insights.

TATABOT — AI SOLUTION BLUEPRINT

USER INPUT

Worker types or speaks a question

"What PPE is needed in furnace area?"

"How do I fix Error Code E-47?"

"What should I learn after induction?"



AI ENGINE

Step 1: Understand user intent & role

Step 2: Search SOPs, manuals, guidelines

Step 3: Generate simple, clear response

Step 4: Personalize by role & experience



OUTPUT

Direct answer in simple language

Step-by-step procedure guidance

Personalized learning recommendation

Action prompt (log issue, next module)

TataBot transforms information into role-specific, real-time guidance



Ramesh

"I am a new operator.
What should I learn first?"



TataBot

- Safety Induction — PPE rules & emergency procedures
- Equipment Familiarization — understand your workstation
- Daily Operational Checklist — tasks before every shift



Personalized learning path generated



Suresh

"Error code E-47 on
the rolling mill.
What does it mean?"



TataBot

- Check motor temperature readings
- Inspect cooling fans and ventilation
- Verify power supply conditions
- Refer Rolling Mill SOP Section 4.2



Instant troubleshooting support

TATABOT IN ACTION — SAMPLE INTERACTIONS (2 of 2)

TataBot



Priya

"Which departments need additional training support?"



TataBot

- Blast Furnace Operations
- Maintenance Team A
- New Operator Cohort

💡 AI-generated workforce insights



Shopfloor Employee

"What safety precautions before operating this equipment?"



TataBot

- PPE requirements checklist
- Equipment inspection steps
- Emergency procedures

💡 Real-time safety guidance



Supervisor

"Show training completion status for my team."



TataBot

- Overall completion percentage
- Pending learners list
- Skill gap indicators

💡 Data-driven learning visibility

MEASURABLE IMPACT — WHY THIS MATTERS

Operational Impact

✓ Query resolution: hours → seconds

✓ Reduced dependency on supervisors

✓ Faster troubleshooting → lower downtime

Learning Impact

✓ Higher training completion rates

✓ Improved knowledge retention

✓ New joiners onboard faster & confidently

Safety Impact

✓ Fewer incidents from knowledge gaps

✓ Improved safety procedure compliance

✓ Real-time access to PPE & LOTO guidance

"Is this usable by a shopfloor worker or only by engineers? — YES, designed for everyone."

WEEK 3

BUILD AI CORE

TataBot: From Design to Working Prototype

Functional Prototype → Agentic AI → Real L&D Queries

"The best way to predict the future is to build it."

WHAT IS AGENTIC AI?

Traditional AI answers one question. Agentic AI takes actions, uses tools, and completes tasks autonomously — just like a smart assistant that thinks step by step.



Perceive

Understands the user's question and context



Think

Decides what information is needed and how to find it



Act

Searches knowledge base, retrieves relevant documents



Respond

Generates a personalized, role-specific answer

TataBot is an Agentic AI — it perceives the user role, thinks about what they need, searches real documents, and responds with personalized guidance.

WEEK 3 — FUNCTIONAL PROTOTYPE BUILT



Python + Groq API

Built TataBot using Python and integrated Groq's free LLM API (Llama 3.3 70B) as the AI brain.



Knowledge Base Connected

Created 3 structured text documents — Safety Procedures, Training Modules, Equipment Troubleshooting.



Q&A Tool Working

TataBot answers real L&D queries in natural language, pulling answers from the knowledge base.



Input → Process → Output

Complete agentic workflow: user asks → TataBot searches → personalized answer generated.



Tested with Real Queries

All 5 core L&D scenarios tested and validated: safety, training, troubleshooting, reporting, boundaries.

Mentor check: "Does it solve the problem or just show AI capability?" → It solves the problem.

WEEK 4

MAKE IT INTELLIGENT

TataBot: From Prototype to Real-World Solution

RAG Implementation → Role Detection → Web Interface

"From demo to deployable — intelligence is in the details."

WHAT IS RAG — RETRIEVAL AUGMENTED GENERATION?

RAG = AI that reads real documents before answering. Instead of relying on training data alone, it retrieves relevant information from your own documents in real time.

✗ WITHOUT RAG (Week 3)

✗ Knowledge hardcoded in prompt

✗ Static — can't update without coding

✗ No source reference

✗ Risk of making up answers

✗ Same answer for everyone

✓ WITH RAG (Week 4)

✓ Reads real .txt documents

✓ Dynamic — update docs, AI updates too

✓ Shows source document used

✓ Validated against real documents

✓ Personalized by user role

WEEK 4 — INTELLIGENT UPGRADES IMPLEMENTED



Real Document Reading

TataBot now reads 3 actual knowledge base files at startup — `safety_procedures.txt`, `training_modules.txt`, `equipment_troubleshooting.txt`



Role Detection Engine

Automatically detects user type (New Operator / Maintenance Technician / L&D Manager) from question keywords and responds accordingly.



Hallucination Prevention

Strict prompt rules prevent TataBot from inventing SOP numbers, procedures, or data not present in documents. Honest 'not found' responses.



Web Interface Built

Full browser-based chat UI using Flask — dark navy theme, role tags, source references, typing indicators, Clear Chat button.



Structured Response Format

Responses use consistent formatting: 📖 Training, ⚠️ Safety, 🔧 Troubleshooting, ⚠️ Warnings, ✓ Actions, Source document cited.

Mentor check: "Would you trust this AI in a real training environment?" → Yes. Source validation + no hallucination.

TATABOT LIVE DEMO — TRAINING GUIDANCE + TROUBLESHOOTING

TB TataBot
AI Knowledge Assistant — Tata Steel Ltd.

00:00:00

Clear Chat

Welcome to TataBot!

I am your AI Knowledge Assistant for Learning and Development.
Ask me about safety, training modules, or equipment troubleshooting.

I am a new operator. What should I learn first?

You

1

RECOMMENDED FOR NEW OPERATOR
📖 TRAINING GUIDANCE
As a new operator, it is essential to start with the basics of safety and equipment familiarization.
✓ Your first step should be to complete Module 1: Safety Induction, which covers:
✓ PPE rules and correct usage
✓ Emergency procedures and evacuation
✓ Workplace hazard awareness
This foundation will help you understand the critical safety aspects of your workplace and ensure your well-being.
📌 Complete Module 1 before proceeding to other modules.
What specific area of the plant will you be working in, so I can provide more tailored guidance?
Source: TRAINING_MODULE1.TXT

I am a maintenance technician. How do I troubleshoot Error Code E-47?

You

2

RECOMMENDED FOR MAINTENANCE TECHNICIAN
🔧 TROUBLESHOOTING GUIDE
ERROR CODE E-47 indicates a Motor Overheating issue.
✓ Symptoms: Motor temperature above safe limit.
To address this, follow these steps:
✓ Stop the machine safely and immediately.
✓ Check motor temperature readings.
✓ Inspect cooling fans and ventilation system.
✓ Verify power supply conditions.
✓ Refer to Rolling Mill SOP Section 4.2 for detailed procedures.
📌 If the issue persists, contact your maintenance supervisor for further assistance.
Remember to check the cooling system before every shift to prevent such issues.
What is the current motor temperature reading, and have you inspected the cooling fans?
Source: EQUIPMENT_TROUBLESHOOTING.TXT

PPE Requirements

New Operator Guide

Error E-47

Training Status

LOTO Procedures

Powered On and Ready to Go

Start

TATABOT LIVE DEMO — SAFETY, REPORTING + KNOWLEDGE BOUNDARY

TB TataBot
AI Knowledge Assistant — Tata Steel Ltd.

• 00:00

Clear Chat

What PPE is required in the furnace area?

You

INFO/KNOWLEDGE SHORTCODE INF00101

SAFETY INFORMATION
When working in the furnace area, it is crucial to wear the required Personal Protective Equipment (PPE) to ensure your safety.
✓ The mandatory PPE for the furnace area includes:
✓ Heat-resistant gloves
✓ Face shield
✓ Fire-resistant jacket
✓ Safety boots
✓ Hard hat
■ Always wear the complete set of required PPE when entering the furnace area to protect yourself from heat and fire hazards.
Are you aware of the location of the nearest fire extinguisher and emergency exit in the furnace area?
Source: SAFETY_PROCEDURES.TXT

Show training completion status for my team

You

INFO/KNOWLEDGE OR L&D SHORTCODE

TRAINING GUIDANCE
As of June 2024, the training completion status for your team is as follows:
✓ Total operators: 120
✓ Completed safety training: 98 (82%)
✓ Pending training: 22 operators
The departments that need support with training are:
✓ Blast Furnace Operations
✓ Maintenance Team A
✓ New Operator Cohort
■ Focus on providing training to the pending 22 operators and supporting the mentioned departments to improve overall training completion rates.
Which specific training module would you like to prioritize for your team, and what support do you need to facilitate the training process?
Source: TRAINING_MODULES.TXT

What is the salary structure at Tata Steel?

You

INFO/KNOWLEDGE OR SHORTCODE INF00101

I could not find this information in the available Tata Steel documents.
Source: None

PPE Requirements

New Operator Guide

Order L-47

Training Status

L&D Procedures

Powered by Gemini 1.5 Pro

Start

WEEK 5

BUSINESS IMPACT

TataBot: Translating Technology into Business Value

ROI Analysis → Scalability → Future Roadmap

"If you cannot explain the value, you have not solved the problem."

BUSINESS CASE — TATABOT ROI & IMPACT ANALYSIS

METRIC	WITHOUT TATABOT	WITH TATABOT	BUSINESS IMPACT
Query Resolution	2-4 hours	< 5 seconds	99% faster
Training Completion	82% (98/120)	95%+ target	+13% improvement
Supervisor Dependency	High — workers wait	Low — instant answers	Freed for critical tasks
Knowledge Loss Risk	High — retirements	Low — captured in docs	Institutional memory saved

Scalability: TataBot can serve all 65,000+ Tata Steel employees simultaneously — 24/7, in any plant, at zero marginal cost per query.

FUTURE ROADMAP — TATABOT V2 & BEYOND

LIVE ✓

Phase 1

Current — MVP

- Terminal + Web chatbot
- 3 knowledge base documents
- Role detection
- Hallucination prevention

PLANNED

Phase 2

3 Months

- Connect to real Tata Steel PDFs
- Voice input for shopfloor workers
- Multi-language support (Hindi)
- Admin dashboard for L&D team

VISION

Phase 3

6 Months

- Integration with HR/LMS systems
- Skill gap auto-assessment
- Personalized learning paths
- Analytics and reporting

Thank You

TataBot — AI Knowledge Assistant for Tata Steel L&D

Problem Identified	Lack of personalized L&D at scale
Solution Built	TataBot — AI Knowledge Assistant
Technology Used	Python, Groq API, Flask, RAG
Impact	99% faster query resolution, scalable to 65,000+ employees

"The right knowledge, to the right employee, at the right time."